Brendan Boyd

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Education

SUNY Stony Brook University 2020-2026
Physics & Astronomy Department
PhD in Physics, with Concentration in Astronomy

Michigan State University 2016-2020
College of Natural Science, Honors College
Bachelor of Science, Astrophysics
Minors in Math and CMSE

Research & Teaching Experience

Graduate Research Assistant – Stony Brook University 2022-Present
Computational Astrophysics Dissertation Research.
Using a low Mach reactive hydrodynamic code to study the convective Urca process in relation to type Ia supernovae.
Multiple publications in academic journals.

Teaching Assistant – Stony Brook University 2020-2021

Undergraduate Research Assistant – Michigan State University 2019-2020
Astrophysics Thesis Research.
Developed SALSA Python package for analyzing galaxy simulations.
Publication in the Journal for Open Source Software (JOSS).

Undergraduate Learning Assistant – Michigan State University 2019-2020

Computational Skills

Programming Skills:
Proficient in Python
Proficient in C++
Competent in FORTRAN
Competent in HPC workflows (e.g. SLURM, bash scripting)
Competent in OpenMP threading
Competent in MPI parallelism

Computational Fluid Modeling
Low Mach Flows
(Nuclear) Reactive Flows

Software Projects - Contributor/Developer

MAESTROeX - github.com/AMReX-Astro/MAESTROeX 2022-Present
pynucastro - github.com/pynucastro/pynucastro 2022-Present
initial models - github.com/AMReX-Astro/initial models 2022-Present
SALSA- github.com/biboyd/SALSA 2019-2021